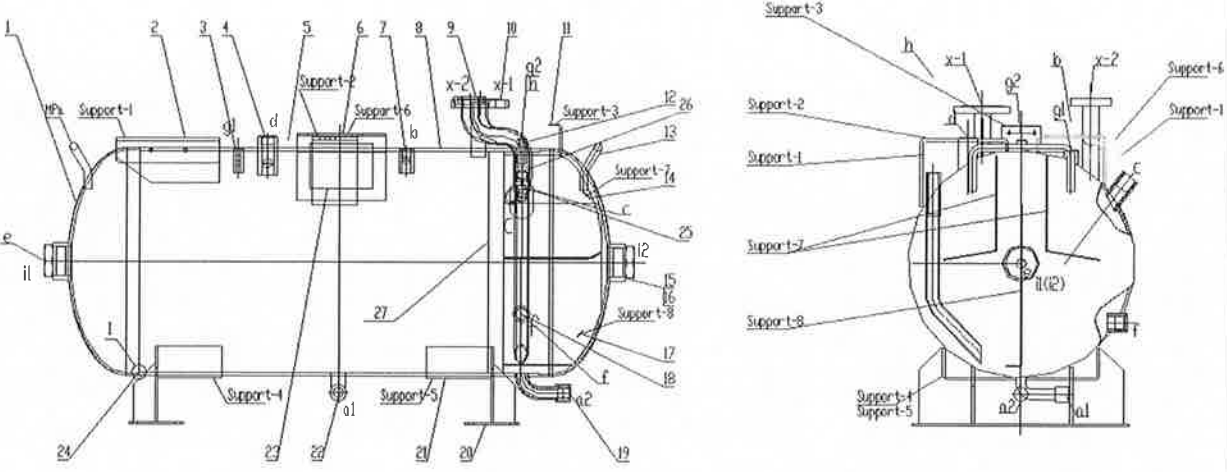


MATERIAL CHECK LIST (A)

材料核对单

Exhibit 4-9A, Edition 3, Revision 0

Doc No.: MCL(A)-P-2362-1 Page 1 of 1

工作令号 JOB No.	P-2362	项目名称 PROJECT NAME	油气分离器 Oil/Gas Separator
图号 DWG. No. 20980145-083-001		修改标记 REV. No.3	
略图 SKETCH			
			
备注 REMARKS: 产品编号 Product No.:P-2362-001~P-2362-025		材料检验员编制 Prepared By 郑如英 2024.1.6 材料工程师审核 Reviewed By 2024.1.6	

MATERIAL CHECK LIST (B)
材料核对单

Exhibit4-9B, Edition 3, Revision 0

Product Drawing NO: 20980145-083-001~015

Doc. No.: MCL(B) -P-2362-1 Page 1 of 1

件号 PART No.	零件名称 PART NAME	材料牌号 MATERIAL SPEC./GRADE.	规格 DIMENSION	数量 QTY	钢厂质保书 Certificate No.	炉批号 HEAT/LOT No.	材料编号 CODED MARKING	质保书类型 Certificate Type
1	Head	SA/GB713Q345R	EHA450*6mm	2	211226P00243	21210811/21210811310302	AC22002	3.1
3	Coupling	SA-105M	RC1/4" Ø25	1	D220312-1	J21137123/B2203121	AY22036	3.2
4	Coupling-d	SA-105M	G3/4" Ø38	1	D220312-1	J21137123/B2203121	AY22042	3.2
7	Coupling-d	SA-105M	Rc3/8" Ø27	1	D220312-1	J21137123/B2203121	AY22038	3.2
8	Shell	SA/GB713Q345R	Ø450*6mm	1	211226P00243	21210811/21210811310302	AC22002	3.1
9-1	Pipe	SA-106 Gr.B	Ø 33.4*4.55	1	20220209-1	852164513/74545616	AP22012	3.1
9-2	Flange	SA/GB713Q345R	T=18	1	210511P00231	21303752/21303752130101	AC22016	3.1
10-1	Pipe	SA-106 Gr.B	Ø 33.4*4.55	1	20220209-1	859544802/74651268	AP22012	3.1
10-2	Flange	SA/GB713Q345R	T=18	1	210511P00231	21303752/21303752130101	AC22016	3.1
12	Coupling	SA-105M	RC1/4" Ø25	1	D220312-1	J21137123/B2203121	AY22036	3.2
15	Plug	SA-105M	G2" Ø80	2	D220312-1	J21137123/B2203121	AY22045	3.2
16	Coupling-1/12	SA-105M	Rp2" Ø80	2	D220312-1	J21137123/B2203121	AY22040	3.2
18	Coupling-f	SA-105M	G3/4" Ø38	1	D220312-1	J21137123/B2203121	AY22042	3.2
19/22-1	Pipe	SA-106 Gr.B	Ø 26.7*3.91	2	20220209-1	859544802/74651268	AP22011	3.1
19/22-2	Coupling-a1/a2	SA-105M	Rc1/2" Ø35	2	D220312-1	J21137123/B2203121	AY22041	3.2
25	Coupling-c	SA-105M	M27*1.5 Ø38	1	D220312-1	J21137123/B2203121	AY22039	3.2
26-1	Pipe	SA-106 Gr.B	Ø 26.7*3.91	1	20220209-1	859544802/74651268	AP22011	3.1
26-2	Coupling-h	SA-105M	G1/2" Ø35	1	D220312-1	J21137123/B2203121	AY22037	3.2
27	Plate	SA/GB713Q345R	T=30	1	210511P00231	21303752/21303752130101	AC22016	3.1

材料工程师审查:

日期:

Reviewed by

Material Engineer:

日期:

材料检验员编制:

Prepared by

Material Examiner :



MATERIAL CHECK LIST (B)
材料核对单

Exhibit4-9B, Edition 3, Revision 0

Product Drawing NO: 20980145-083-016~025

Doc. No.: MCL(B) -P-2362-2 Page 1 of 1

件号 PART No.	零件名称 PART NAME	材料牌号 MATERIAL SPEC./GRADE.	规格 DIMENSION	数量 QTY	钢厂质保书 Certificate No.	炉批号 HEAT/LOT No.	材料编号 CODED MARKING	质保书类型 Certificate Type
1	Head	SA/GB713Q345R	EHA450*6mm	2	211226P00243	21210811/21210811310302	AC22002	3.1
3	Coupling	SA-105M	RC1/4" Ø25	1	D220312-1	J21137123/B2203121	AY22036	3.2
4	Coupling-d	SA-105M	G3/4" Ø38	1	D220312-1	J21137123/B2203121	AY22042	3.2
7	Coupling-d	SA-105M	Rc3/8" Ø27	1	D220312-1	J21137123/B2203121	AY22038	3.2
8	Shell	SA/GB713Q345R	Ø450*6mm	1	211226P00243	21210811/21210811310302	AC22002	3.1
9-1	Pipe	SA-106 Gr.B	Ø 33.4*4.55	1	20220209-1	852164513/74545616	AP22012	3.1
9-2	Flange	SA/GB713Q345R	T=18	1	210511P00231	21303752/21303752130101	AC22016	3.1
10-1	Pipe	SA-106 Gr.B	Ø 33.4*4.55	1	20220209-1	859544802/74651268	AP22012	3.1
10-2	Flange	SA/GB713Q345R	T=18	1	210511P00231	21303752/21303752130101	AC22016	3.1
12	Coupling	SA-105M	RC1/4" Ø25	1	D220312-1	J21137123/B2203121	AY22036	3.2
15	Plug	SA-105M	G2" Ø80	2	D220312-1	J21137123/B2203121	AY22045	3.2
16	Coupling-i1/i2	SA-105M	Rp2" Ø80	2	D220312-1	J21137123/B2203121	AY22040	3.2
18	Coupling-f	SA-105M	G3/4" Ø38	1	D220312-1	J21137123/B2203121	AY22042	3.2
19/22-1	Pipe	SA-106 Gr.B	Ø 26.7*3.91	2	20220209-1	859544802/74651268	AP22011	3.1
19/22-2	Coupling-a1/a2	SA-105M	Rc1/2" Ø35	2	D220312-1	J21137123/B2203121	AY22041	3.2
25	Coupling-c	SA-105M	M27*1.5 Ø38	1	D220312-1	J21137123/B2203121	AY22039	3.2
26-1	Pipe	SA-106 Gr.B	Ø 26.7*3.91	1	20220209-1	859544802/74651268	AP22011	3.1
26-2	Coupling-h	SA-105M	G1/2" Ø35	1	D220312-1	J21137123/B2203121	AY22037	3.2
27	Plate	SA/GB713Q345R	T=30	1	210511P00231	21303752/21303752130101	AC22016	3.1

材料检验员编制:

郑长英

日期:

2024.1.16

材料工程师审查:

郑长英

日期:

2024.1.16

Prepared by

郑长英

Reviewed by

郑长英

Material Examiner :

Material Engineer:

Date:





MILL TEST CERTIFICATE

热轧钢板
Prime Hot-rolled Steel plate
锅炉和压力容器用钢板

车船号码	TRAIN/SHIP NO
苏A2A049	

化学成分

CHEMICAL ANALYSIS (wt%)		
Cr	Ni	Mo
100	0	0
99.9	0.1	0
99.8	0.2	0
99.7	0.3	0
99.6	0.4	0
99.5	0.5	0
99.4	0.6	0
99.3	0.7	0
99.2	0.8	0
99.1	0.9	0
99.0	1.0	0
98.9	1.1	0
98.8	1.2	0
98.7	1.3	0
98.6	1.4	0
98.5	1.5	0
98.4	1.6	0
98.3	1.7	0
98.2	1.8	0
98.1	1.9	0
98.0	2.0	0
97.9	2.1	0
97.8	2.2	0
97.7	2.3	0
97.6	2.4	0
97.5	2.5	0
97.4	2.6	0
97.3	2.7	0
97.2	2.8	0
97.1	2.9	0
97.0	3.0	0
96.9	3.1	0
96.8	3.2	0
96.7	3.3	0
96.6	3.4	0
96.5	3.5	0
96.4	3.6	0
96.3	3.7	0
96.2	3.8	0
96.1	3.9	0
96.0	4.0	0
95.9	4.1	0
95.8	4.2	0
95.7	4.3	0
95.6	4.4	0
95.5	4.5	0
95.4	4.6	0
95.3	4.7	0
95.2	4.8	0
95.1	4.9	0
95.0	5.0	0
94.9	5.1	0
94.8	5.2	0
94.7	5.3	0
94.6	5.4	0
94.5	5.5	0
94.4	5.6	0
94.3	5.7	0
94.2	5.8	0
94.1	5.9	0
94.0	6.0	0
93.9	6.1	0
93.8	6.2	0
93.7	6.3	0
93.6	6.4	0
93.5	6.5	0
93.4	6.6	0
93.3	6.7	0
93.2	6.8	0
93.1	6.9	0
93.0	7.0	0
92.9	7.1	0
92.8	7.2	0
92.7	7.3	0
92.6	7.4	0
92.5	7.5	0
92.4	7.6	0
92.3	7.7	0
92.2	7.8	0
92.1	7.9	0
92.0	8.0	0
91.9	8.1	0
91.8	8.2	0
91.7	8.3	0
91.6	8.4	0
91.5	8.5	0
91.4	8.6	0
91.3	8.7	0
91.2	8.8	0
91.1	8.9	0
91.0	9.0	0
90.9	9.1	0
90.8	9.2	0
90.7	9.3	0
90.6	9.4	0
90.5	9.5	0
90.4	9.6	0
90.3	9.7	0
90.2	9.8	0
90.1	9.9	0
90.0	10.0	0
89.9	10.1	0
89.8	10.2	0
89.7	10.3	0
89.6	10.4	0
89.5	10.5	0
89.4	10.6	0
89.3	10.7	0
89.2	10.8	0
89.1	10.9	0
89.0	11.0	0
88.9	11.1	0
88.8	11.2	0
88.7	11.3	0
88.6	11.4	0
88.5	11.5	0
88.4	11.6	0
88.3	11.7	0
88.2	11.8	0
88.1	11.9	0
88.0	12.0	0
87.9	12	

χ_{10}^2	χ_{10}^2	χ_{10}^2	χ_{10}^2
---------------	---------------	---------------	---------------

DATE	BY	TO	AMOUNT	BALANCE
5	3	0.37	31	

5	3	0.3	31
---	---	-----	----

5	3	0.3	31
5	3	0.3	31

5	3	0,3	31
5	3	0,3	31

2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----

WTAH Certifies that this material meets the requirements of SA/B 5713 Q345FR

17

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----

[illegible]

1000

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----

100

臨

五

二

★

4

卷之五

100

NO-DIMENSION CHECK-OK

人 員 投 資 部

PLS MOL234567	80137-90137-90
---------------	----------------

一、

2

原证证明书为准。Note: Grade: 0

证书编号 Certificate No. D220312-1										建湖县旭盛石化机械有限公司 Jianhu Country Xusheng Petrochemical Co., Ltd										客户名称: 安徽威尔泰克机电设备有限公司 Purchaser: Anhui weier teck mechanical and electrical equipment co.,LTD																																																																																																													
材料名称 Name of Material: 锻件 Forging										Jianhu Country Xusheng Petrochemical Co., Ltd										用户工作令号 Customer Work No.: P-2201-P-2210																																																																																																													
材料标准 Mat.Spec.: ASME SA-105/SA-105M-2021										产品质量证明书 QUALITY CERTIFICATE										报告日期 Date of Report: 2022/03/12																																																																																																													
锻件标准 Standard SA-105M										级别 Grade II										锻造比 Forging Rate ≥3										热处理状态 Heat Treatment										正火 Normalizing										批号 Batch No. B2203121																																																																															
化学成分 Chemical Composition %																																																																																																																																	
牌号 Heat No.																																																																																																																																	
C										Si										Mn										S										P										Cr										Ni										Mo										Cu										V										Nb										N										Co									
标准值 S.D ≤0.35										0.10-0.35										0.60-1.05										≤0.040										≤0.035										≤0.30										≤0.40										≤0.12										≤0.40										0.08										/										/										/									
冶炼值 H.A 0.159										0.27										1.30										0.007										0.017										0.040										0.016										0.002										0.045										0.003										/										/										/									
试样位置 Sample										试样 Sample										探伤检查 Detection check										符合标准及级别 Acceptance Standard & Class										其他试验 Other Test										试验或评定标准 Test or Evaluation Standard										试验结果 Test Result																																																																					
单独锻制试样 A Separately Forged Piece Ø30×200																																																																																																																																	
机械性能试验数据 Mechanical Property																																																																																																																																	
拉伸试验 Tensile Test																																																																																																																																	
试样规格 Size of Sample										试验温度 Test Temp.										屈服强度 T.S. σ _s (MPa)										抗拉强度 T.S. σ _b (MPa)										断面收缩率 RA %										试验温度 Test Temp.										试样规格 Size of Sample										缺口形式 Type of Notch										冲击功 Akv(J) ≥27J										硬度 Hardness(HB)																																							
直径 Dia(mm)										室温 Room Temp.										≥485										≥250										≥22										≥30										0℃										/										/										95										102										107										≤187									
12.5										50										525										350										35										70										0℃										/										/										95										102										107										≤187									
产品规格 Detailed Product Specification																																																																																																																																	
序号 Item No.										图号/Standard No. Dwg./Standard No.										名称 Name										规格 Size										数量 Qty.										检验尺寸 Dim. Inspection										外观检查 Visual Inspection										结论 Result										备注 Remarks																																																	
1										20980145-083-002										Thread pie										RC1/4 Φ25*40										500										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
2										20980145-083-002										Thread pie										G1/2 Φ35*90										200										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
3										20980145-083-002										Thread pie										RC3/8Φ27*40										200										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
4										20980145-083-002										Thread pie										M27*1.5Φ38*75										200										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
5										20980145-083-002										Thread pie										RP2Φ80*40										200										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
6										20245137-010-002										Thread pie										RC1/2Φ35*110										300										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
7										20245137-006-002										Thread pie										G3/4 Φ38*125										400										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
8										20245137-006-002										Thread pie										NPT1/2Φ35*110										100										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
9										/										Round bar										Φ16*1000										20										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
10										/										Round bar										Φ80*500										20										Acceptable 合格										Acceptable 合格										Acceptable 合格										K																																																	
注: 兹证明, 本合格证中的材料、产品是按上述标准和技术条件进行制造、取样、试验和检验的, 并满足要求。We hereby certify that the material and technical specification and satisfied with the requirement.																																																																																																																																	
Note: manufactured, sampled, inspected and tested in accordance with the above standard and technical specification and satisfied with the requirement.																																																																																																																																	
检验员: 祖许明										Inspector: 祖许明										WTAAH Certifies that this material meets the requirements of the Code Section II 2021 and (Edition) (Addenda)										SGS WITNESSED REVIEWED NOTED FOR INFORMATION ONLY C SGS-CSC STANDARDS TECHNICAL SERVICE CO., LTD 20220323										制造单位 Manufacturer: 梅徐梅																																																																																									



Inspection Certificate (EN 10204-3.2)

Customer: Inspection No.: IN-SH-5805-21377
Certificate No.: IN-SH-5805-23177-02
Order No.: /

Well- Tanker (Anhui) M&E Equipment Co.Ltd.

Date: /

Manufacturer:

Works No.: N/A

Jianhu Country Xusheng Petrochemical Co.,Ltd

Article: forging

Technical specification/Requirements: ASME SA-105 ED2021

Material: SA-105
according to: ASME SA-105

Edition: 2021

State of Delivery: Normalizing

Melting Process: N/A

Typical Marking:

Brand of the manufacture: N/A

Grade, heat No., Size

Inspector's name and stamp: Jerry Chi IND023

Extent of material delivery:

Item No.	Number of pieces	Article	Heat No./Batch No.	Remark
1	500 (pcs)	RC1/4 Φ25*40	J21137123/ B2203121	
	200 (pcs)	G1/2 Φ35*90		
	200 (pcs)	RC3/8Φ27*40		
	200 (pcs)	M27*1.5Φ38*75		
	200 (pcs)	RP2Φ80*40		
	300 (pcs)	RC1/2Φ35*110		
	400 (pcs)	G3/4 Φ38*125		
	100 (pcs)	NPT1/2Φ35*110		
	20 (pcs)	Φ 16*1000		
	20 (pcs)	Φ 80*500		

Additional remarks: NA

The requirements are fulfilled as per annex:

Location and Date: Hefei City, Jun. 17, 2022



This document is issued by the Company under its General Conditions of Service accessible at http://www.sgs.com/terms_and_conditions.rtf. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein.
Any other holder of this document is advised that information could be used between the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and its document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law.

[V-02, Sep 29th, 2017]

Test results:
Annex: For IN-SH-5805-21337

Mechanical tests

Specimen type	State of delivery	Specimen No.	Dim. Of specimen	Test Specimen	Yield strength	Tensile strength	Elongation	Red are of	Values	1 = Energy of impact [J] 2 = Impact strength [J/cm ²] 3 = Cryst. Proportion [%] 4 = Expansion [mm.10 ⁻²] 5 = Expansion [%] 6 = Hardness [HB] 7 = Bend test [d=1.5a, 180°]
J21137123/ B2203121		Ø (mm)	Thick (mm)	°C	R _p (MPa)	R _m (MPa)	A _{Lo=50}	Z (%)	6 152 2 144 1 102 3 108 2 152 155	OK
J21137123/ B2203121		12.5	/	/	26	351	522.9	34	71.5	OK
Note: impact test temperature: 0 °C, specimen size: 10*10*55mm										
Chemical analyse(%)										
Melt No.	C	SI	P	S	Mn	Cu	Cr	Mo	NI	V
J21137123/ B2203121	0.157	0.274	0.017	0.007	1.31	0.045	0.039	0.001	0.016	0.002

We certify that the materials listed above have met the requirements of ASME SA-105 ED2021.

Material manufacturer:
Jianhu Country Xusheng Petrochemical
Co., Ltd

Signature and stamp



Date:

Date:

2021 19

This document is issued by the Company under its General Conditions of Service accessible at http://www.ogs.com/terms_and_conditions.html. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein.

Any other holder of this document is advised that information contained herein reflects the Company's findings at the time of its intervention only and to the limits of the Company's sole responsibility as its Client, and this document does not exonerate parties to transaction from exercising all their rights and obligations under the transaction documents. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law.

[V:02, Sep. 29th, 2017]

Page 2 of 2



天长市康弘石油管材有限公司
TIANCHANG KANGHONG OIL PIPE COMPANY LIMITED
产 品 质 量 证 明 书
MILL TEST CERTIFICATE

WTAB Certifies that this material meets the requirements of the Code Section II 201 and 202.4.0
Sign: [Signature] Date: 2022.5.25

产品标准及名称: ASME SA106/SA106M-2019 高温用无缝碳钢公称管
Product Standard & Name: EN10204 3.1
热处理制度及交货状态: 正火
热处理: 正火
Voucher No: 20220209-1

序号 No.	炉号 Heat No.	批号 Lot No.	化学成分 (Chemical Composition) %										力学性能 Mechanical Properties		金相组织 Metallography		无损检测 Non-destructive Testing		数量 (Quantity)	
			C	Si	Mn	P	S	Cr	Ni	Cu	Ti	Mo	V	Al	屈服强度 Yield Strength	抗拉强度 Tensile Strength	伸长率 Elongation	断面收缩率 Reduction of Area	屈服强度 Yield Strength	抗拉强度 Tensile Strength
1	SA106Gr. B	AP22-011	26.70±0.91												屈服强度 Yield Strength	抗拉强度 Tensile Strength	伸长率 Elongation	断面收缩率 Reduction of Area	屈服强度 Yield Strength	抗拉强度 Tensile Strength
2	SA106Gr. B	AP22-012	33.40±0.55												屈服强度 Yield Strength	抗拉强度 Tensile Strength	伸长率 Elongation	断面收缩率 Reduction of Area	屈服强度 Yield Strength	抗拉强度 Tensile Strength

1. 表面质量、外形尺寸均符合产品技术要求。
2. 有争议时, 以证书编号、批次、规格和重量。
3. 生产许可证编号。
1. The Surface quality, outer dimension all meet the requirements technical.
2. If dispute arise, please tell us detail of certification No., Lot, No. Size & Weight.
3. Production license code.

说明
Note

4. 我们特此证明, 所有产品均符合客户要求, 取样、试验和检验, 并符合合同要求。
4. We hereby certify that all products have been manufactured, sampled, tested and inspected in accordance to and in compliance with the requirements of the contract and Customer Purchase Order.

认证机构: ABSG Consulting Inc.
认证日期: 2022.5.25
Date of Issue: 2022.5.25
电话 (TEL): 0550-6518858 传真 (FAX): 0550-7513111

天中市康弘石油管材有限公司

TIANCHANG KANGHONG OIL PIPE COMPANY LIMITED

产品质量证明书

MILL TEST CERTIFICATE

EN10204 3.1

产品标准及名称: ASME SA106/SA106M-2019 高温用无缝碳钢管

热处理制度及交货状态: 正火

序列号: 2021071101-2

Witness
Certifies that this
certified copies are true copies of
of the (Order Section) of 200 and 1
Sign 康弘 Date 2021.7.5
(Signature) (Address)



Product Standard & Name				Chemical Composition (%)												Mechanical Properties		Heat Treatment		Non-destructive Test		Inspection	
Order No.	Grade	Spec Size (mm)	Lot No.	Heat No.	C	Si	Mn	P	S	Cr	Ni	Cu	Mo	Al	Y	U.T.	Yield Strength	U.T.	U.T.	U.T.	U.T.	U.T.	U.T.
1	SA106Gr B	22*2.77	9875360	21578926	21	22	52	14	8	0.001	4-0.0001	5-0.00001	Y	Y	Y	Y	530	530	530	530	530	530	530
2	SA106Gr B	26*4.56	9875301	21497057	21	22	51	14	8	0.001	4-0.0001	5-0.00001	Y	Y	Y	Y	530	530	530	530	530	530	530

1. 表面质量: 外形尺寸应符合产品技术要求。
2. 化学成分: 符合证书编号: 批次、规格和重量。
3. 生产许可证编号:
1. The Surface quality, outer dimension all meet the requirements technical.
2. If dissonance arise, please tell us detail of Certification No., Lot, No. Size & Weight.
3. Production license code:
说明
Note
4. 按此证明此系列产品, 均按照标准以及订单的要求制造, 取样, 试验和检验, 并符合合同要求。
4. We hereby certify that the products described herein have been manufactured, sampled, tested and inspected in accordance with the requirements of the contract and Customer Purchase Order.
证书签发人: 康弘
Certification Signature: Kang Hong
签发日期: 20210711 10:01
Date of Issue: 2021.7.11
电话 (TEL): 0550-7513111
邮编 (PC): 239011
地址: 安徽省天长市康弘石油管材有限公司
Address: Zhenjiang Rd, Tongcheng Tianchang Oil Pipe Co., Ltd.

PMA Particular Material Appraisal

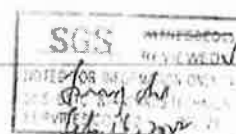
PMA Ref. No.		PMA-SA/GB713 Q345R	
Pressure equipment Manufacturer:		Well-Tanker(Anhui) M&E Equipment Co., Ltd.	
Material specification:	Designation / Grade	Delivery Condition:	
ASME Sec.II	SA/GB713 Q345R	AS-rolled or Nonnormalized	
Material group of welding as per ISO/TR 15608:2017	ISO/TR 15608 Group 1.2		
Application:		All fluids and mixtures in the temperature ranges where corrosion resistance is recognised, by technical literature, to be good or excellent.	
Thickness range:		6-60mm	
Applicable Design Code:		ASME Sec.VIII 1 2021	
Max. allowable temperature (°C):	200	Min. allowable temperature (°C):	-20
Property	Requirement	Compliance	
Appropriate properties (Annex 1 § 4.1 (a))	e.g. R_{eH} $R_{m/20}$ if required R_{mH}	<i>Thickness 6~16mm</i> <i>Tensile strength : 510~640MPa</i> <i>Yield strength: ≥ 345MPa ,</i> <i>Thickness >16~36mm</i> <i>Tensile strength : 500~630MPa</i> <i>Yield strength: ≥ 325MPa for</i> <i>Thickness >36~60mm</i> <i>Tensile strength : 490~620MPa</i> <i>Yield strength: ≥ 315MPa</i>	
Sufficiently ductile (Annex 1 § 4.1 (a)) (Annex 1 § 7.5)	In steel: A_{min} 14%	$A_{min} \geq 25\%$ for gauge length	
Sufficiently tough (Annex 1 § 4.1 (a), Annex 1 § 7.5)	e.g. in steel: 27J at 20°C or lowest operating temp.	It shall be required in the purchasing specification that materials shall be impact tested at temperature not higher than minimum design temperature, absorbed energy not less than 27J.	
Not significantly affected by ageing (Annex 1 § 4.1 (c))	At intended operating temperatures	Yes	
Suitable for intended processing (Annex 1 § 4.1 (d))	e.g. welding forming etc.	Yes	
Material Marking (Annex 1 § 3.1.5)	Identifiable by Certificate	Marking requirement as per material specification	
Type of Certificate (Annex 1 § 4.3)	Cat. I and non main pressure bearing parts: Certificate of Compliance	Material certificate is required.	
	Cat. II, III, IV (main pressure bearing parts)	■ 3.1 (EN 10204 : 2004)	■ 3.2 (EN 10204 : 2004)
Material Manufacturers Approval	Certified Quality System by organisation established in the EC with specific product control or NoBo to witness test	■	■

Additional testing results are given in Test report No.

Restrictions: NA

Additional Requirements:

Signature: (Manufacturer)	Date:
Approved by Notified Body (Cat. III / IV): Identification Number : 1155	Date of approval:



PMA Particular Material Appraisal

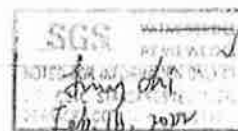
PMA Ref. No	PMA-SA-105		
Pressure equipment Manufacturer:	Well-Tanker(Anhui) M&E Equipment Co., Ltd.		
Material specification: SA-105 2021	Designation / Grade: NA	Delivery Condition: Normalized	
Material group of welding as per ISO/TR 15608:2017	ISO/TR 15608 Group 11.1		
Application:	All fluids and mixtures in the temperature ranges where corrosion resistance is recognised, by technical literature, to be good or excellent.		
Thickness range:	< 200mm		
Applicable Design Code:	ASME SEC. VIII, DIV.1 2021		
Max. allowable temperature (°C):	200	Min. allowable temperature (°C):	-20
Property	Requirement	Compliance	
Appropriate properties (Annex 1 § 4.1 (a))	e.g. R_{eH} , R_{m20} if required R_{mH}	Tensile strength: $\geq 485 \text{ MPa}$ Yield strength: $\geq 250 \text{ MPa}$	
Sufficiently ductile (Annex 1 § 4.1 (a)) (Annex 1 § 7.5)	In steel: A_{min} 14%	$A_{min} \geq 22\%$	
Sufficiently tough (Annex 1 § 4.1 (a), Annex 1 § 7.5)	e.g. in steel: 27J at 20°C or lowest operating temp.	Not specified in SA-105. It shall be required in the purchasing specification that materials shall be impact tested at temperature not higher than minimum design temperature, absorbed energy not less than 27J.	
Not significantly affected by ageing (Annex 1 § 4.1 (c))	At intended operating temperatures	Yes	
Suitable for intended processing (Annex 1 § 4.1 (d))	e.g. welding forming etc.	Yes	
Material Marking (Annex 1 § 3.1.5)	Identifiable by Certificate	Marking requirement as per material specification	
Type of Certificate (Annex 1 § 4.3)	Cat. I and non main pressure bearing parts: Certificate of Compliance Cat. II, III, IV (main pressure bearing parts)	Material certificate is required.	
Material Manufacturers Approval	Certified Quality System by organisation established in the EC with specific product control or	■ 3.1 (EN 10204 : 2004)	■ 3.2 (EN 10204 : 2004)
	NoBo to witness test	■	■

Additional testing results are given in Test report No.

Restrictions:

Additional Requirements:

Signature: (Manufacturer)	Date:
 Approved by Notified Body (Cat. III / IV): Identification Number : 1155	Date of approval:



PMA Particular Material Appraisal

PMA Ref. No.		PMA-SA-106 Gr B	
Pressure equipment Manufacturer:		Well-Tanker(Anhui) M&E Equipment Co., Ltd.	
Material specification:	Designation / Grade:	Delivery Condition:	
SA-106	Gr. B	Hot drawn or Cold drawn with heat treated at a temperature of 650°C or higher.	
Material group of welding as per ISO/TR 15608:2017	ISO/TR 15608 Group 11.1		
Application:		All fluids and mixtures in the temperature ranges where corrosion resistance is recognised, by technical literature, to be good or excellent.	
Thickness range:		≤20mm	
Applicable Design Code:		ASME SEC. VIII, DIV. 1 2021	
Max. allowable temperature (°C):	200	Min. allowable temperature (°C):	-20
Property	Requirement	Compliance	
Appropriate properties (Annex 1 § 4.1 (a))	e.g. R_{eH} $R_{m/20}$ if required R_{mH}	Tensile strength : $\geq 415\text{MPa}$ Yield strength: $\geq 240\text{MPa}$	
Sufficiently ductile (Annex 1 § 4.1 (a)) (Annex 1 § 7.5)	In steel: A_{min} 14%	Elongation shall conform to Note A of Table 2 of SA-106M and be at least 14%.	
Sufficiently tough (Annex 1 § 4.1 (a), Annex 1 § 7.5)	e.g. in steel: 27J at 20°C or lowest operating temp.	Not specified in SA-106. It shall be required in the purchasing specification that materials shall be impact tested at temperature not higher than minimum design temperature, absorbed energy not less than 27J.	
Not significantly affected by ageing (Annex 1 § 4.1 (c))	At intended operating temperatures	Yes	
Suitable for intended processing (Annex 1 § 4.1 (d))	e.g. welding forming etc.	Yes	
Material Marking (Annex 1 § 3.1.5)	Identifiable by Certificate	Marking requirement as per material specification	
Type of Certificate (Annex 1 § 4.3)	Cat. I and non main pressure bearing parts: Certificate of Compliance Cat. II, III, IV (main pressure bearing parts)	Material certificate is required.	
Material Manufacturers Approval	Certified Quality System by organisation established in the EC with specific product control or	■ 3.1 (EN 10204 : 2004)	■ 3.2 (EN 10204 : 2004)
	NoBo to witness test	■	■

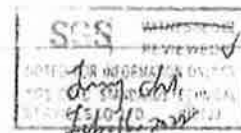
Additional testing results are given in Test report No.

Restrictions:

(a) Impact test may be waived when test specimen at least 5mm wide cannot be obtained.

Additional Requirements:

Signature: (Manufacturer)	Date:
Approved by Notified Body (Cat. III / IV): Identification Number : 1155	Date of approval:





EU CERTIFICATE OF QUALITY SYSTEM APPROVAL

In accordance with the requirements of the Pressure Equipment Directive
2014/68/EU

This is to certify that the Quality Management System of:

Nanjing Iron & Steel Co., Ltd.
Xiejadian, Liuhe District, Nanjing City, Jiangsu Province,
People's Republic of China

has been assessed against the requirements of Annex I, Paragraph 4.3 of the Pressure
Equipment Directive 2014/68/EU and EN 764-5: 2014, Section 4.2, and conforms to the
requirements for the products shown below:

Hot rolled steel plates and bars for mechanical and pressure purposes in accordance
with Annex I paragraph 4.3 of the Pressure Equipment Directive
(see attached schedule)

Approval is subject to the continued maintenance of the quality system in accordance with
the requirements of the above Directive and EN 764-5: 2014, Section 4.2, and the on-going
surveillance. Products are as detailed on the attached schedule.

Certificate No.:	0343/PEDM/BJG0061012/A
Original Approval:	24 October 2008
Current Certificate:	9 March 2023
Certificate Expiry:	8 March 2026
Notified Body No.	0343

Alfred Yao on behalf of LRQA Nederland B.V.

LRQA Nederland B.V. (Reg. no. 24247948) is a private limited company registered in the Netherlands with registered office at George Hintzenweg 77, 3068 AX Rotterdam. A subsidiary of LRQA Group Limited.

LRQA Group Limited, its affiliates and subsidiaries and their respective officers, employees or agents are, individually and collectively, referred to in this clause as 'LRQA'. LRQA assumes no responsibility and shall not be liable to any person for any loss, damage or expense caused by reliance on the information or advice in this document or howsoever provided, unless that person has signed a contract with the relevant LRQA entity for the provision of this information or advice and in that case any responsibility or liability is exclusively on the terms and conditions set out in that contract.



CERTIFICATE SCHEDULE
0343/PEDM/BJG0061012/A

In accordance with the requirements of the Pressure Equipment Directive
2014/68/EU

Nanjing Iron & Steel Co., Ltd.
Xiejadian, Liuhe District, Nanjing City, Jiangsu Province,
People's Republic of China

Product(s): Hot rolled steel plates

Standards	Grades
ASME SA542/SA542M	GrD CL3, GrD CL4 (4.5- 180mm)
ASME SA553/SA553M	Type1 (4.5-50mm)
ASME SA662/SA662M	GrA, GrB, GrC (4.5-80mm)
ASME SA737/SA737M	GrB, GrC (4.5-80mm)
JIS G3103	SB410, SB450, SB450M, SB480, SB480M (4.5-80mm)
JIS G 3115	SPV235, SPV315, SPV355, SPV410, SPV450, SPV490 (4.5-80mm)
JIS G 3118	SGV410, SGV450, SGV480 (4.5-80mm)
JIS G 3124	SEV245, SEV295, SEV345 (4.5-80mm)
JIS G 3126	SLA365, SLA325A, SLA325B, SLA235A, SLA235B, SLV235A (4.5-80mm)
GB713	Q245R, Q345R (6-200mm) 14Cr1MoR, 12Cr2Mo1R, 12Cr2Mo1VR (6-180mm), Q370R, 15CrMoR (6-100mm)
GB3531	16MnDR, 15MnNiDR, 09MnNiDR, 08Ni3DR (6-100mm)
GB19189	12MnNiVR (4.5-60mm) 07MnNiMoVDR, 07MnCrMoVR (12-60mm)

Schedule Issue: 03
Date of Schedule Issue: 9 March 2023
Notified Body No. 0343


Alfred Yao on behalf of LRQA Nederland B.V.

LRQA Nederland B.V. (Reg. no. 24247948) is a private limited company registered in the Netherlands with registered office at George Hintzenweg 77, 3068 AX Rotterdam. A subsidiary of LRQA Group Limited.
LRQA Group Limited, its affiliates and subsidiaries and their respective officers, employees or agents are, individually and collectively, referred to in this clause as 'LRQA'. LRQA assumes no responsibility and shall not be liable to any person for any loss, damage or expense caused by reliance on the information or advice in this document or howsoever provided, unless that person has signed a contract with the relevant LRQA entity for the provision of this information or advice and in that case any responsibility or liability is exclusively on the terms and conditions set out in that contract.



CERTIFICATE

Certificate No.
10.16.1358

Date of initial Issue
06/08/2019

Valid From
15/06/2023

The Certificate is valid until
05/08/2024

Tianchang Kanghong Oil pipe Co., Ltd.
Zhenxing Road, Tongcheng Town, Tianchang City, Anhui
Province, China.

This is to certify that, upon the relevant request of the applicant, EUROCERT as Third Party Authority has certified that the above manufacturer implements, operates and maintains a Quality Assurance System in accordance with the Pressure Equipment Directive 2014/68/EU (PED) Annex I, Paragraph 4.3 as Material Manufacturer for the scope of manufacture of Seamless steel tube of carbon steel, alloy steel.

Relevant products covered by this certificate are listed in ANNEX A to this certificate.

The manufacturer is, therefore, authorized to issue certificates of specific product control within the scope of the assessed quality system and in accordance with the Pressure Equipment Directive 2014/68/EU and applied technical specifications.

Possible additional requirements-specific to applied technical specifications to meet PED Annex I are not affected.

The validity of this certificate is subject to a valid certification in accordance with ISO 9001:2015 with a related to production of materials for pressure equipment scope.

This certificate remains valid only when the organisation passes the annual surveillance audits.



Check the validity of the Certificate by scanning the QR code at right
EUROCERT S.A. 89 Chlois & Lykavrisseos str., 144 52, Metamorphosi - Greece
T +30 210 62.52.495, +30 210 62.53.927, F +30 210 62.03.018, M eurocert@otenet.gr



ANNEX TO

CERTIFICATE NO.: 10.16.1358

LIST OF MATERIAL

Specification	Grade	Specification	Grade
ASTM A53/A53M ASME SA53/SA53M	A/ B	GB/T3087	10, 20
ASTM A106/A106M ASME SA106/SA106M	A/ B/ C	GB/T5310	20G, 20MnG, 25MnG, 12CrMoG, 15CrMoG, 12Cr2MoG, 12Cr1MoVG
ASTM A179/A179M ASME SA179/SA179M	A179/ SA179	EN10216-1	P235TR1, P265TR1, P235TR2, P265TR2
ASTM A192/A192M ASME SA192/SA192M	A192/ SA192	EN10216-2	P235GH, P265GH, 16Mo3, 10CrMo9-10, 13CrMo4-5
ASTM A210/A210M ASME SA210/SA210M	A-1, C	JIS G3454	STPG370, STPGG410
ASTM A213/A213M ASME SA213/SA213M	T2, T5, T11, T12, T22	JIS G3462	STBA12, STBA22
ASTM A333/A333M ASME SA333/SA333M	Gr.1, Gr.6	JIS G3461	STB340, STB410, STB510
ASTM A334/A334M ASME SA334/SA334M	Gr.1, Gr.6	API5CT	H40 PSL1, J55 PSL1, K55 PSL1, N80(1) PSL1
ASTM A335/A335M ASME SA335/SA335M	P2, P5, P11, P12, P22	API5L	PSL1: L245/B, L290/X42, L320/X46, L360/X52 PSL2: L245R/BR, L290R/X42R, L245N/BN, L290N/X42N, L320N/X46N, L360N/X52N
GB/T8163	20, Q345A, Q345B, Q345C, Q345D, Q345E		
DIN17175	St35.8, St45.8, 15Mo3, 13 CrMo44, 10CrMo910	DIN1629	ST37.0, ST44.0, ST52

Eurocert in providing services does not warrant the accuracy of any information or service supplied by Eurocert, its employees, or its representatives except as set out in the valid certificate and it is not liable for any loss, damage, or expense whatever sustained by any person due to any act or omission or error of whatsoever nature and howsoever caused by Eurocert. The only exception is when any person using Eurocert's services or relying on any information given by and on behalf of Eurocert and suffers loss or expense thereby which is proved to have been due to act, omission or error of Eurocert only then Eurocert will pay compensation to such person that will not in any case exceed the amount of the fee charged by Eurocert for that particular service.



Check the validity of the Certificate by scanning the QR code at right.
EUROCERT S.A. 89 Chlois & Lykovriseos str., 144 52, Metamorphosi - Greece
T +30 210 62.52.495, +30 210 62.53.927, F +30 210 62.03.018, M eurocert@otenet.gr

